

Multi-Agent Path Planning in GPS-Denied Environments with Localization Constraints and Optimality Guarantees

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Abstract—Centralized multi-agent path planning with cooperative localization (MAPP-CL) in GPS-denied environments is generally computationally intractable when formulated in full belief space due to tightly coupled uncertainty dynamics. Prior work addresses this challenge through scalable approximate belief-space methods that sacrifice optimality or formulations that provide guarantees only under restrictive modeling assumptions. We study a multi-agent planning problem in which a primary agent must reach a goal in minimum time while satisfying a terminal constraint on localization uncertainty. We produce a tractable surrogate of the underlying belief-space problem by optimizing over the mean-state trajectories of all agents and propagating covariance for terminal constraint evaluation. We propose a discrete-to-continuous pipeline combining hexagonal-grid joint search with spline-based trajectory refinement. The resulting solution is ϵ -suboptimal with respect to the optimal feasible solution in the continuous state space. Empirically, the approach improves upon baselines by placeholder% and reduces computation time by placeholder%, while satisfying terminal uncertainty constraints.

I. INTRODUCTION

In environments where GPS signal is unreliable, such as underwater or in regions subject to spoofing or jamming, GPS-denied solutions must be employed. Autonomous multi-agent systems operating in such environments suffer from state uncertainty caused by inertial measurement unit (IMU) drift. As estimation error accumulates over time, an agent’s pose uncertainty can grow to the point where waypoint tracking degrades and mission objectives are compromised. Common planning strategies mitigate this growth by biasing trajectories toward informative landmarks and maintaining inter-agent measurements to enable cooperative localization. Consequently, trajectory planning must account for uncertainty propagation when estimation error significantly impacts task performance. While belief-space planning provides a principled formulation for uncertainty-aware decision-making, it is typically intractable in multi-agent settings due to the dimensionality of coupled belief dynamics. This motivates surrogate formulations that optimize mean trajectories while propagating uncertainty only for constraint evaluation.

In this work, we consider a centralized multi-agent planning problem in which a primary agent must reach a goal in minimum time under constant-speed motion, which is equivalent to minimizing path length. The objective is subject to a terminal constraint on localization uncertainty.

We compute a coordinated discrete plan over a joint state space that captures motion constraints and inter-agent ranging capabilities. This solution is then lifted into continuous space and refined into smooth, dynamically feasible

trajectories that satisfy curvature constraints and terminal uncertainty requirements.

Our contributions are as follows:

- 1) A centralized multi-agent planning strategy for GPS-denied environments that explicitly couples task execution with uncertainty-aware support, enabling optimal performance of a primary agent under bounded localization uncertainty.
- 2) A scalable discrete-to-continuous planning pipeline programmed in Julia, combining hexagonal-grid joint search with spline-based trajectory refinement and yielding ϵ -suboptimal solutions with respect to the optimal continuous-time feasible trajectory.
- 3) *Placeholder* A comprehensive empirical evaluation in HoloOcean, demonstrating the effectiveness of the proposed approach in reducing primary-agent path length while maintaining localization reliability across multi-agent GPS-denied scenarios.

In the following sections, we formalize the problem, describe the discrete and continuous planning stages in detail, and evaluate our approach in simulated GPS-denied environments with multiple supporting agents.

II. RELATED WORK

Prior work on belief-space planning can be broadly categorized into two classes. Approximate belief-space methods improve scalability by simplifying uncertainty propagation or optimizing surrogate objectives, but generally sacrifice global optimality guarantees [1], [2]. In contrast, methods that retain stronger theoretical guarantees typically rely on restrictive assumptions such as full observability, limited coupling between agents, or simplified noise models.

Multi-robot planning methods range from centralized to decoupled. Centralized joint belief-space planning can in principle yield globally optimal solutions but suffers from exponential growth of the state space. Conflict-Based Search (CBS) is a well-known centralized MAPP solver that splits conflicts between agent paths to improve efficiency [3], though it operates in a purely discrete setting without uncertainty. In contrast, decoupled and prioritized planners plan independently for each robot and resolve conflicts or coordinate online, improving scalability but often ignoring dynamical coupling or estimation effects. Distributed schemes and formation controllers are also used in cooperative localization to maintain structured geometries that reduce drift, but these heuristics generally lack formal guarantees on global optimality.

Grid-based and sampling-based planners provide practical tools for handling large environments under kinematic constraints. Hexagonal grid discretizations reduce directional bias and provide uniform neighbor spacing, leading to more isotropic path representations [4]. We adopt a 2D hex grid for the discrete search to better approximate curvature constraints and reduce quantization error. After a discrete solution is obtained, continuous refinement enforces dynamic feasibility. Spline-based trajectory representations, particularly B-splines, are widely used due to their smoothness and local control. Recent works integrate B-splines into sampling-based planners to satisfy curvature limits; for example, variants of RRT* generate tree expansions composed of curvature-constrained B-spline segments [5], [6]. These methods embody a hybrid discrete-to-continuous paradigm, combining global search with smooth trajectory optimization.

Cooperative localization and uncertainty propagation are typically handled using distributed estimation frameworks. Kalman filters and their variants propagate agent state covariances, while covariance intersection (CI) enables consistent fusion when cross-correlations are unknown [7]. Consensus-based filters allow agents to iteratively exchange estimates with neighbors [8]. Extensions often incorporate adaptive or distance-dependent weighting of inter-agent updates to account for communication quality and information relevance. In particular, spatially decaying Gaussian weights are commonly used to model how sensing and communication efficacy diminish with distance [2]. These principles motivate our use of distance-weighted Kalman updates during planning to bound the growth of localization uncertainty under communication constraints.

III. PROBLEM FORMULATION

Let $\mathcal{I} = \{1, 2, \dots, P\}$ denote a team of P agents. Agent $i = 1$ is the primary agent tasked with reaching a goal location $G \in \mathbb{R}^2$. All agents move at constant speed v_i , making minimization of time equivalent to minimizing trajectory length.

A. Decision Variables

The optimization is performed over mean-state trajectories. Specifically, the decision variables are the heading sequences

$$\{\theta_i[k]\}_{i \in \mathcal{I}, k=0}^{N-1}.$$

Positions are deterministically induced by these controls:

$$x_i[k+1] = x_i[k] + v_i \Delta t \begin{bmatrix} \cos \theta_i[k] \\ \sin \theta_i[k] \end{bmatrix}.$$

B. Covariance Propagation

Let $\Sigma_i[k]$ denote the covariance of agent i . Covariance is not a decision variable and is propagated deterministically for constraint evaluation.

We assume linearized motion with anisotropic process noise:

$$\Sigma_i[k+1|k] = F(\theta_i[k]) \Sigma_i[k|k-1] F(\theta_i[k])^\top + Q_i[k],$$

where

$$Q_i[k] = \begin{bmatrix} \sigma_{\parallel}^2[k] + \alpha_{\parallel} \Delta t & 0 \\ 0 & \sigma_{\perp}^2[k] + \alpha_{\perp} \Delta t \end{bmatrix}.$$

C. Measurement Model (Tapered Gaussian Sensing)

When an agent is within sensing range of a landmark or another agent, a measurement update is applied. The Kalman gain is

$$K_i[k] = \Sigma_i[k|k-1] H^\top (H \Sigma_i[k|k-1] H^\top + R)^{-1},$$

and the covariance is updated using the Joseph form:

$$\Sigma_i[k] = (I - K_i[k] H) \Sigma_i[k|k-1] (I - K_i[k] H)^\top + K_i[k] R K_i[k]^\top.$$

Sensing quality is modeled via a tapered Gaussian function of distance:

$$w_{ij}(k) = \exp \left(-\frac{\|x_i[k] - x_j[k]\|^2}{2\sigma_c^2} \right),$$

which modulates measurement availability.

D. Constraints

Terminal uncertainty constraint:

$$\det(\Sigma_i[N]) \leq \bar{d}, \quad \forall i \in \mathcal{I}.$$

Here, $\bar{d}$ is the maximum allowable terminal covariance determinant.

Dynamics:

$$x_i[k+1] = x_i[k] + v_i \Delta t \begin{bmatrix} \cos \theta_i[k] \\ \sin \theta_i[k] \end{bmatrix}.$$

Curvature constraint:

$$|\theta_i[k+1] - \theta_i[k]| \leq \frac{v_i \Delta t}{r}.$$

Initial conditions:

$$x_i[0] = x_i^0, \quad \Sigma_i[0] = \Sigma_i^0, \quad \forall i \in \mathcal{I}.$$

E. Optimization Problem

$$\min_{\{\theta_i[k]\}, N} N$$

subject to the above constraints.

F. Optimal Continuous Solution

Let L^* denote the optimal feasible solution cost in the continuous state space, defined as the minimum-time trajectory satisfying all motion, curvature, and terminal uncertainty constraints under exact continuous dynamics.

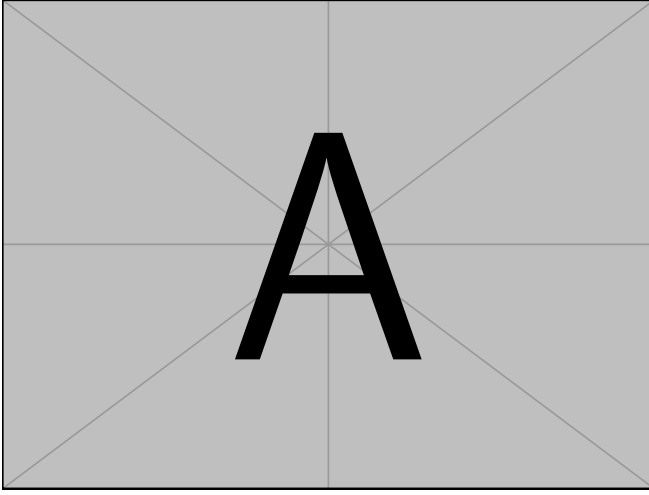


Fig. 1. Discrete joint A* solutions over a coupled multi-agent graph, later refined in continuous space.

IV. GLOBAL DISCRETE SEARCH

We first solve the problem on a joint discrete graph using standard A* over the composite state of all agents. Let $\mathbf{x} = (x_1, \dots, x_P)$ denote a joint state and let $g(\mathbf{x})$ denote the accumulated primary path cost from the start. The evaluation function is

$$f(\mathbf{x}) = g(\mathbf{x}) + h(\mathbf{x}), \quad (1)$$

where $h(\mathbf{x})$ is an admissible heuristic defined using only the primary agent's state. Specifically, $h(\mathbf{x})$ is computed as a shortest-path lower bound on the underlying graph, obtained via precomputed all-pairs shortest paths. Support agents contribute zero heuristic cost.

To improve search efficiency, we apply sound dominance pruning among states with the same joint node tuple. Given two states A and B at the same joint tuple, A dominates B if

$$g(A) \leq g(B) \quad \text{and} \quad \det(\Sigma_A) \leq \det(\Sigma_B),$$

where Σ_A and Σ_B denote the primary-agent covariance matrices at the corresponding states. If one state dominates the other under this criterion, the dominated state is discarded. This pruning is sound for our objective because the heuristic is shared at the same node tuple, the primary path cost of the dominated state cannot be lower, and the covariance metric used for feasibility assessment cannot improve under future propagation.

Under standard assumptions of admissible and consistent heuristics and nonnegative edge costs, A* returns an optimal feasible solution on the pruned discrete joint graph. Let L^* denote the optimal feasible trajectory cost in the continuous state space defined above. Let L_{returned} denote the cost of the trajectory returned by the proposed pipeline. This optimality holds with respect to the discrete graph cost.

We assume the discrete graph is a geometric discretization of the continuous environment with resolution parameter h . Under this discretization, there exists a constant $\epsilon_{\text{disc}} \geq 0$

such that for any continuous feasible trajectory with cost L^* , there exists a discrete feasible trajectory satisfying

$$L_{\text{disc}}^* \leq (1 + \epsilon_{\text{disc}}) L^*. \quad (2)$$

This captures the approximation error induced by finite graph resolution and path quantization.

V. REFINING CONTINUOUS TRAJECTORIES

The discrete solution is lifted into continuous space and refined using clamped cubic B-splines. Let P_{disc} denote the discrete path obtained from the joint A* search, and let $R(\cdot)$ denote the spline refinement operator that maps P_{disc} to a continuous trajectory τ_{spline} .

We define:

$$\tau_{\text{spline}} = \mathcal{R}(P_{\text{disc}}). \quad (3)$$

The refinement operator enforces C^2 continuity and curvature feasibility constraints, while producing a smooth trajectory that interpolates or approximates the discrete path. However, this transformation introduces geometric distortion relative to the original discrete solution.

We model this distortion as an additional multiplicative suboptimality term:

$$L(\tau_{\text{spline}}) \leq (1 + \epsilon_{\text{spline}}) L(P_{\text{disc}}), \quad (4)$$

where $\epsilon_{\text{spline}} \geq 0$ captures the approximation error induced by spline smoothing, curvature enforcement, and deviation from piecewise-linear geodesics.

Unlike the discrete solution, spline refinement is not cost-preserving in general, as curvature constraints and smoothness requirements may increase trajectory length relative to the discrete path.

VI. GUARANTEES AND BOUNDS

Proposition 1 (End-to-end ϵ -suboptimality). *Assume: (i) A* returns an optimal solution on the discrete graph, (ii) the discretization induces a bounded approximation error ϵ_{disc} such that any continuous feasible trajectory admits a discrete approximation satisfying*

$$L_{\text{disc}} \leq (1 + \epsilon_{\text{disc}}) L^*, \quad (5)$$

and (iii) spline refinement induces bounded distortion ϵ_{spline} such that

$$L_{\text{spline}} \leq (1 + \epsilon_{\text{spline}}) L_{\text{disc}}. \quad (6)$$

Then the returned trajectory satisfies:

$$L_{\text{returned}} \leq (1 + \epsilon_{\text{disc}})(1 + \epsilon_{\text{spline}}) L^*. \quad (7)$$

Equivalently, for small $\epsilon_{\text{disc}}, \epsilon_{\text{spline}}$, we obtain:

$$L_{\text{returned}} \leq (1 + \epsilon_{\text{total}}) L^*, \quad \epsilon_{\text{total}} \approx \epsilon_{\text{disc}} + \epsilon_{\text{spline}}. \quad (8)$$

Spline refinement introduces additional suboptimality due to curvature enforcement and geometric smoothing, and should be interpreted as a bounded approximation operator rather than a cost-preserving projection.

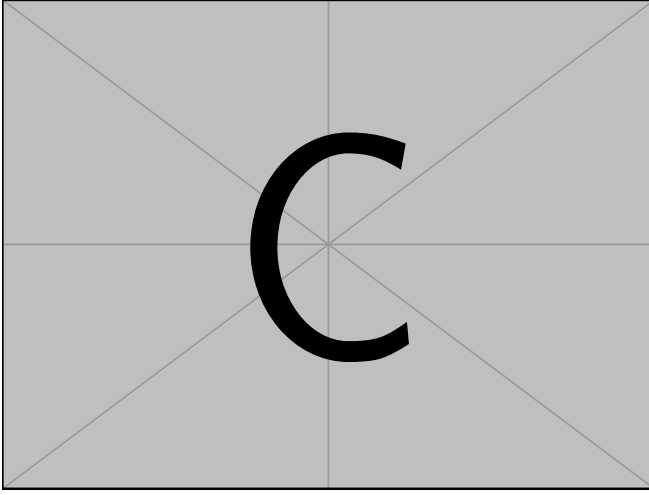


Fig. 2. Figure showing Holocean simulator results over several trials compared with baselines.

VII. LIMITATIONS AND ASSUMPTIONS

Despite the theoretical structure of the proposed pipeline, several limitations remain.

First, the end-to-end optimality guarantee is conditional on discretization and refinement assumptions. In particular, the ϵ -suboptimality result relies on the existence of a sufficiently fine discretization of the continuous space and a bounded distortion property of the spline refinement. If either assumption is violated (e.g., coarse graphs or aggressive curvature constraints), the bound may not accurately reflect performance.

Second, the continuous refinement stage is a local non-linear projection procedure that enforces curvature and feasibility constraints. While it does not worsen trajectory cost under the admissible refinement operator, its final outcome may depend on initialization from the discrete solution.

Third, uncertainty propagation is based on an information-form approximation that neglects higher-order correlations and unknown cross-covariances between agents. While this improves computational tractability, it may introduce conservatism or inconsistency in tightly coupled sensing scenarios.

Fourth, feasibility enforcement relies on penalty and barrier methods rather than hard constraint satisfaction throughout optimization. Although constraints are enforced at convergence and during candidate evaluation, intermediate iterates may temporarily violate constraints.

Finally, the approach assumes that the underlying graph structure and motion primitives sufficiently capture the connectivity of the continuous environment. In highly cluttered or dynamically changing environments, this assumption may fail, leading to degraded solution quality or loss of feasible paths.

VIII. RESULTS

IX. CONCLUSION AND FUTURE WORK

Unifying paragraph on approach and results

Future work on speeding up pipeline MPC/POMCP for online planning, improved surrogate modeling of the belief space, removing bspline approximation error by using bsplines in the discrete space, etc.

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APPENDIX ACKNOWLEDGMENT